

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Copper oxides based compounds have been widely studied in recent decades, due to their peculiar phase diagram and the wide observed phenomenology. Among the various phases, these materials exhibit long-range antiferromagnetic ordering when undoped, and high-temperature anisotropic superconductivity when charge is injected. The latter, when discovered, was particularly unexpected: indeed, at normal state these materials show an insulating phase. It has been understood, however, that the insulation mechanism is incompatible with conventional band theory; it results, instead, from the strong on-site Coulomb repulsion between electrons. Namely, these materials are Mott insulators. Both the antiferromagnetic and superconducting phase are deeply connected to the physics of Mott insulation.

Most of the low-energy properties of copper oxides are well described by the Hubbard model on bidimensional square lattice. The latter is composed of two parts: the kinetic operator, representing the hopping of electrons on neighbouring site, and the on-site repulsion between electrons of opposite spin, which originates from Coulomb interaction. The Hubbard model can host Mott insulation, antiferromagnetism and superconductivity, and advanced numerical techniques lead in general to very good compatibility with many experiments.

In recent years, it has been proposed that a more accurate description of the low-energy spectrum of copper oxides could have been reached by adding a new interaction to the Hubbard model, namely a non-local attraction between electrons on neighbouring sites. The resulting model has been denominated “extended Hubbard model”.

[Motivations: from *d*-wave to Mott...]

In this thesis we study the extended Hubbard model using the Hartree-Fock method. Namely, we implement a constrained variational search in a subspace of the many-body Hilbert space, and approximate the thermal state through the best many-body state inside the selected subspace. In the Hartree-Fock method, the variational search is limited to the set of all states that are the ground-state or some non-interacting fermionic hamiltonian.

The Hartree-Fock method provides a simple framework to study competition among phases. Each competing phase is described by the spontaneous breaking of one or more symmetries of the extended Hubbard model. Hence, by constraining the variational search to the subspace of states that show the symmetries of a given phase, we can obtain the best approximation to the true ground-state inside that subspace. Then, comparing the free energies of the resulting approximations, we can assess which phase of those studied is energetically favoured.

We study competition among three phases: normal, antiferromagnetic and superconducting.

[To be continued...]